



# COMP47350: Data Analytics (Conv)

## Lecture 9: Feature Pairs

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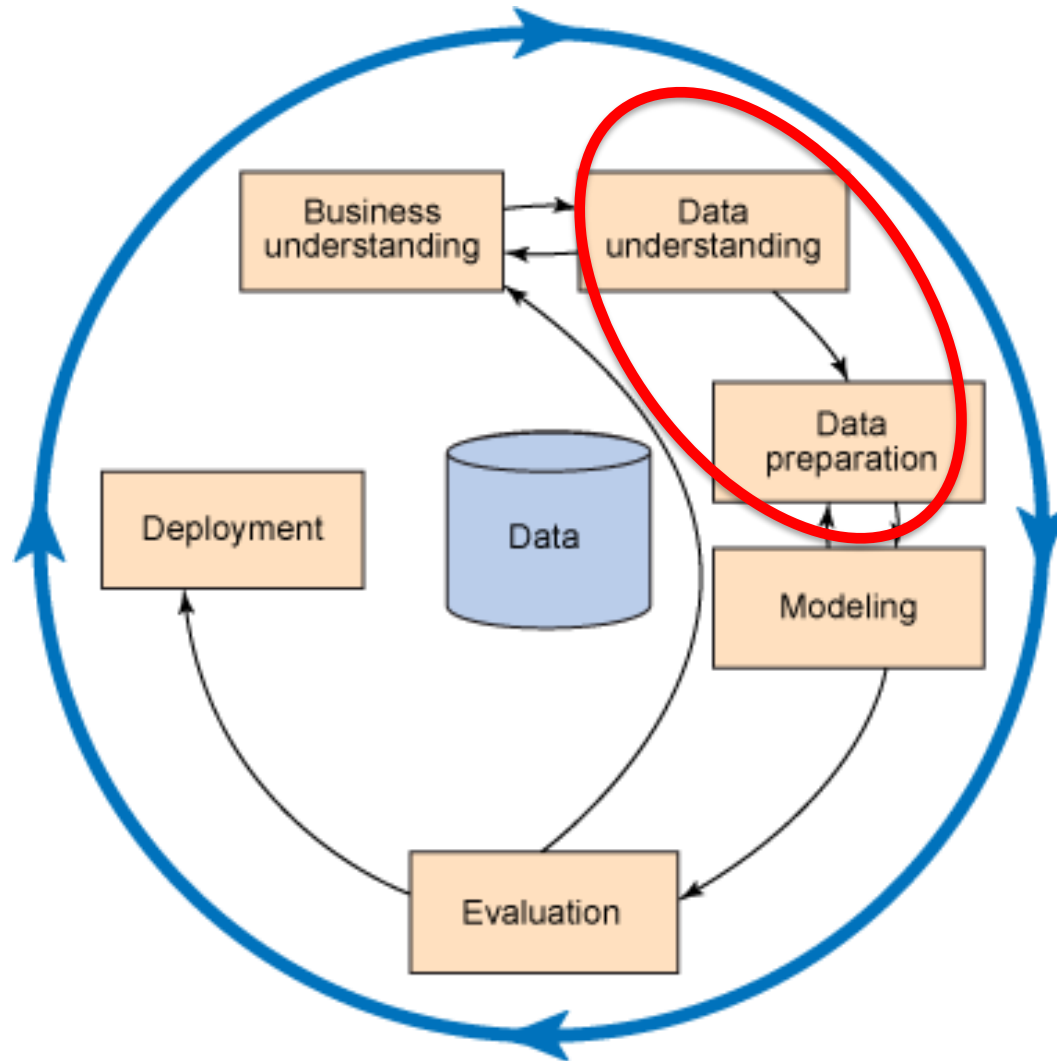
# Module Topics

- **Python Environment** (Anaconda, Jupyter Notebook)
- **Getting Data** (Web scrapping, APIs, DBs)
- **Understanding Data** (slicing, visualisation)
- **Preparing Data** (cleaning, transformation)
- **Modeling & Evaluation** (machine learning)

# Data Analytics Project Lifecycle:

## CRISP-DM

CRISP-DM: **C**Ross-**I**ndustry **S**tandard **P**rocess for **D**ata **M**ining



# Data Understanding: Recap

## 1. Data Quality Report: Analyze each feature

- Tables for descriptive stats
- Plots for data distribution
- Discussion of findings (short written report)

## 2. Data Quality Plan: Prepare each feature

- Problems due to invalid or valid data
  - Potential solutions (action plan to be implemented)
- We often need to re-create the tables and plots once the features are cleaned
  - At the end of data preparation (DQR + DQP) there should be no more NaN values (i.e., either replace them or drop them). **The actions in the DQP aim to clean the data.**

# Data Quality Plan

- Focus so far was on analyzing and preparing **single features**
  - Why isn't that enough?
  - Why do we care about how two features interact?
- We often need to analyze **feature pairs**, to understand **feature correlation** and to check if a given feature is predictive of the target outcome
  - *Find patterns, causality, or redundancy*

# Data Understanding: Feature Pairs

**This lecture:**

- **Visualizing Relationships Between Feature-Pairs through Plots (plots)**
  - Continuous vs continuous
  - Categorical vs categorical
  - Continuous vs categorical
- **Measuring correlation between features (numeric measure)**

# Basketball ABT

| ID | POSITION | HEIGHT | WEIGHT | CAREER<br>STAGE | AGE | SPONSORSHIP<br>EARNINGS | SHOE<br>SPONSOR |
|----|----------|--------|--------|-----------------|-----|-------------------------|-----------------|
| 1  | forward  | 192    | 218    | veteran         | 29  | 561                     | yes             |
| 2  | center   | 218    | 251    | mid-career      | 35  | 60                      | no              |
| 3  | forward  | 197    | 221    | rookie          | 22  | 1,312                   | no              |
| 4  | forward  | 192    | 219    | rookie          | 22  | 1,359                   | no              |
| 5  | forward  | 198    | 223    | veteran         | 29  | 362                     | yes             |
| 6  | guard    | 166    | 188    | rookie          | 21  | 1,536                   | yes             |
| 7  | forward  | 195    | 221    | veteran         | 25  | 694                     | no              |
| 8  | guard    | 182    | 199    | rookie          | 21  | 1,678                   | yes             |
| 9  | guard    | 189    | 199    | mid-career      | 27  | 385                     | yes             |
| 10 | forward  | 205    | 232    | rookie          | 24  | 1,416                   | no              |
| 11 | center   | 206    | 246    | mid-career      | 29  | 314                     | no              |
| 12 | guard    | 185    | 207    | rookie          | 23  | 1,497                   | yes             |
| 13 | guard    | 172    | 183    | rookie          | 24  | 1,383                   | yes             |
| 14 | guard    | 169    | 183    | rookie          | 24  | 1,034                   | yes             |
| 15 | guard    | 185    | 197    | mid-career      | 29  | 178                     | yes             |
| 16 | forward  | 215    | 232    | mid-career      | 30  | 434                     | no              |
| 17 | guard    | 158    | 184    | veteran         | 29  | 162                     | yes             |
| 18 | guard    | 190    | 207    | mid-career      | 27  | 648                     | yes             |
| 19 | center   | 195    | 235    | mid-career      | 28  | 481                     | no              |
| 20 | guard    | 192    | 200    | mid-career      | 32  | 427                     | yes             |
| 21 | forward  | 202    | 220    | mid-career      | 31  | 542                     | no              |
| 22 | forward  | 184    | 213    | mid-career      | 32  | 12                      | no              |
| 23 | forward  | 190    | 215    | rookie          | 22  | 1,179                   | no              |
| 24 | guard    | 178    | 193    | rookie          | 21  | 1,078                   | no              |
| 25 | guard    | 185    | 200    | mid-career      | 31  | 213                     | yes             |
| 26 | forward  | 191    | 218    | rookie          | 19  | 1,855                   | no              |
| 27 | center   | 196    | 235    | veteran         | 32  | 47                      | no              |
| 28 | forward  | 198    | 221    | rookie          | 22  | 1,409                   | no              |
| 29 | center   | 207    | 247    | veteran         | 27  | 1,065                   | no              |
| 30 | center   | 201    | 244    | mid-career      | 25  | 1,111                   | yes             |

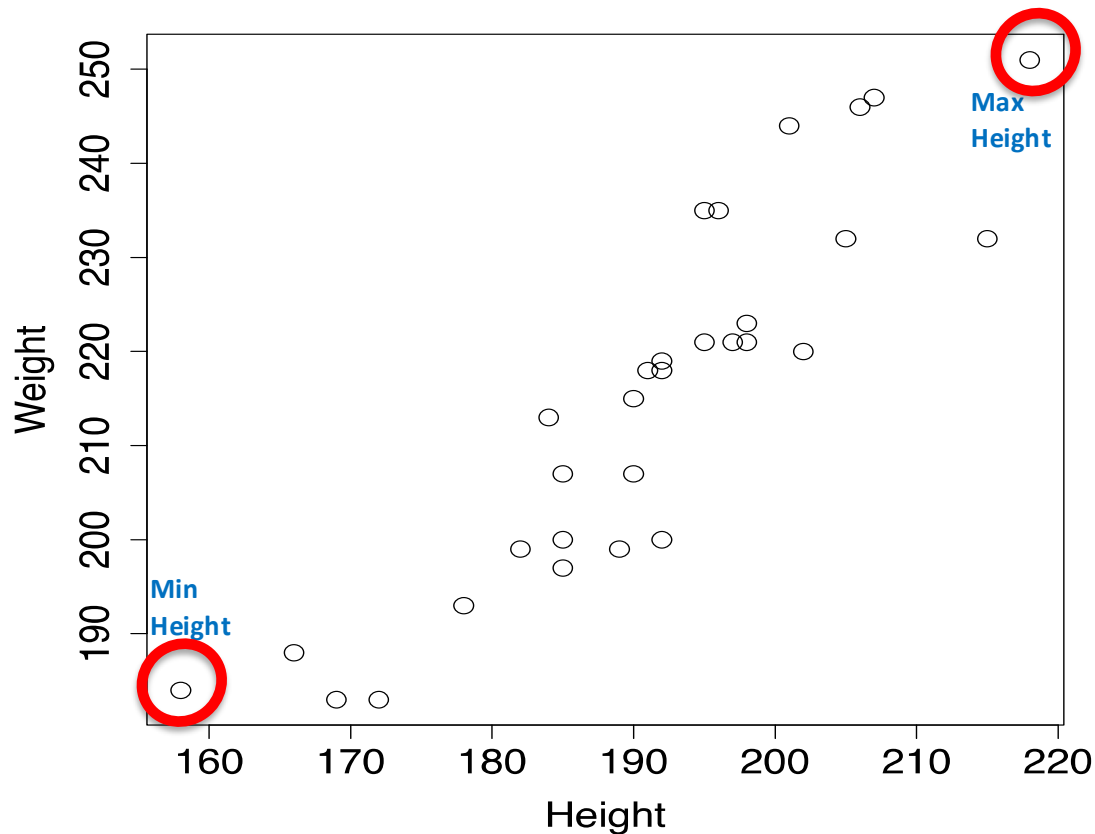
# Visualizing Relationships Between Features

- **Scatter plot** = Plot to visualize **pairs of continuous features** (continuous feature vs continuous feature)
- The horizontal axis (X-axis) represents one feature, and the vertical axis (Y-axis) represents a second feature
- A dot in the plot is an instance (row) in the ABT, where we only plot 2 features (2 columns together)



# Continuous vs Continuous: Scatter Plot

Features: **Height** (X-axis) and **Weight** (Y-axis)  
are positively correlated

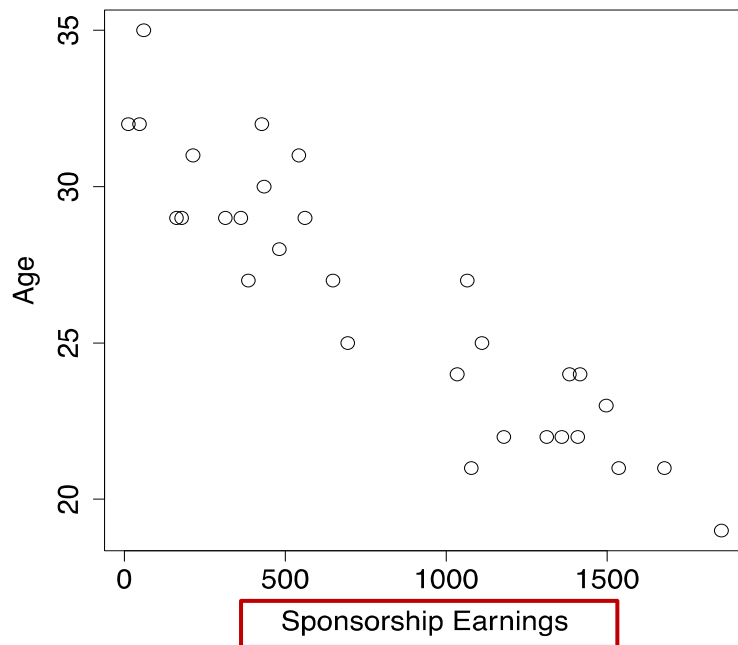


|            | HEIGHT | WEIGHT |
|------------|--------|--------|
| Max Height | 192    | 218    |
|            | 218    | 251    |
|            | 197    | 221    |
|            | 192    | 219    |
|            | 198    | 223    |
|            | 166    | 188    |
|            | 195    | 221    |
|            | 182    | 199    |
|            | 189    | 199    |
|            | 205    | 232    |
| Min Height | 206    | 246    |
|            | 185    | 207    |
|            | 172    | 183    |
|            | 169    | 183    |
|            | 185    | 197    |
|            | 215    | 232    |
|            | 158    | 184    |
|            | 190    | 207    |
|            | 195    | 235    |
|            | 192    | 200    |
|            | 202    | 220    |
|            | 184    | 213    |
|            | 190    | 215    |
|            | 178    | 193    |
|            | 185    | 200    |
|            | 191    | 218    |
|            | 196    | 235    |
|            | 198    | 221    |
|            | 207    | 247    |
|            | 201    | 244    |

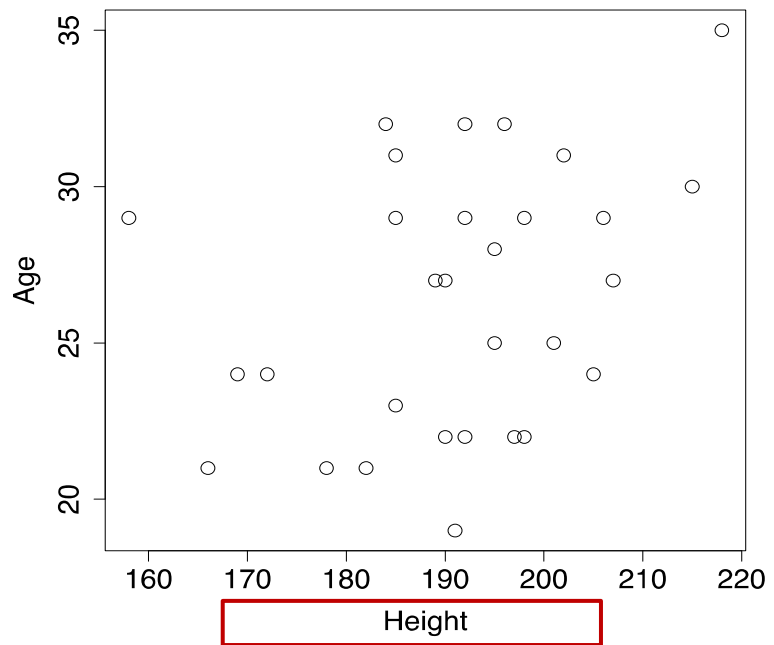
# More Scatter Plots

(a) **Sponsorship Earnings vs Age** (left side; negatively correlated)

(b) **Height vs Age** (right side; no strong correlation)



(a)

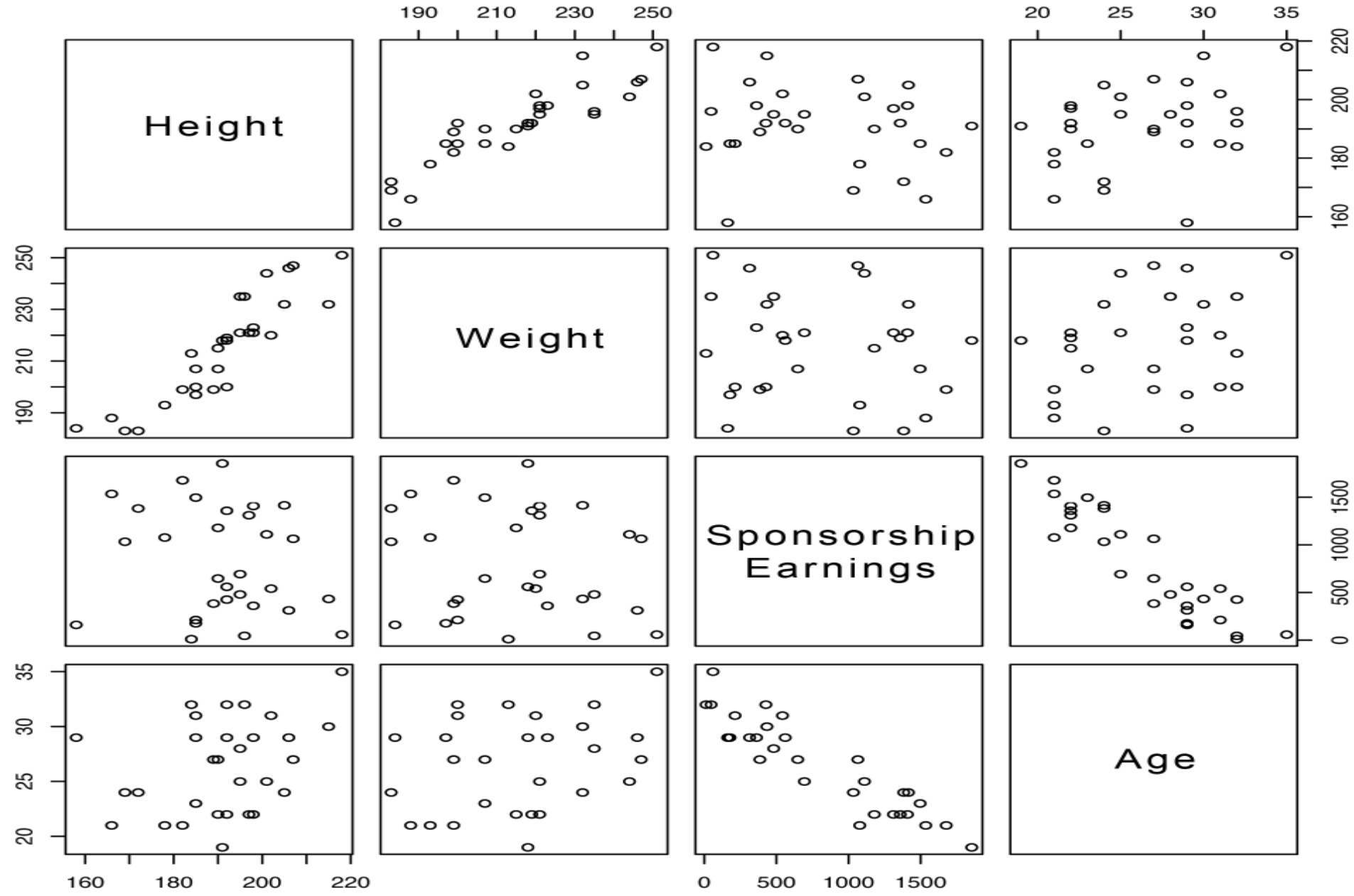


(b)

# Scatter Plot Matrix

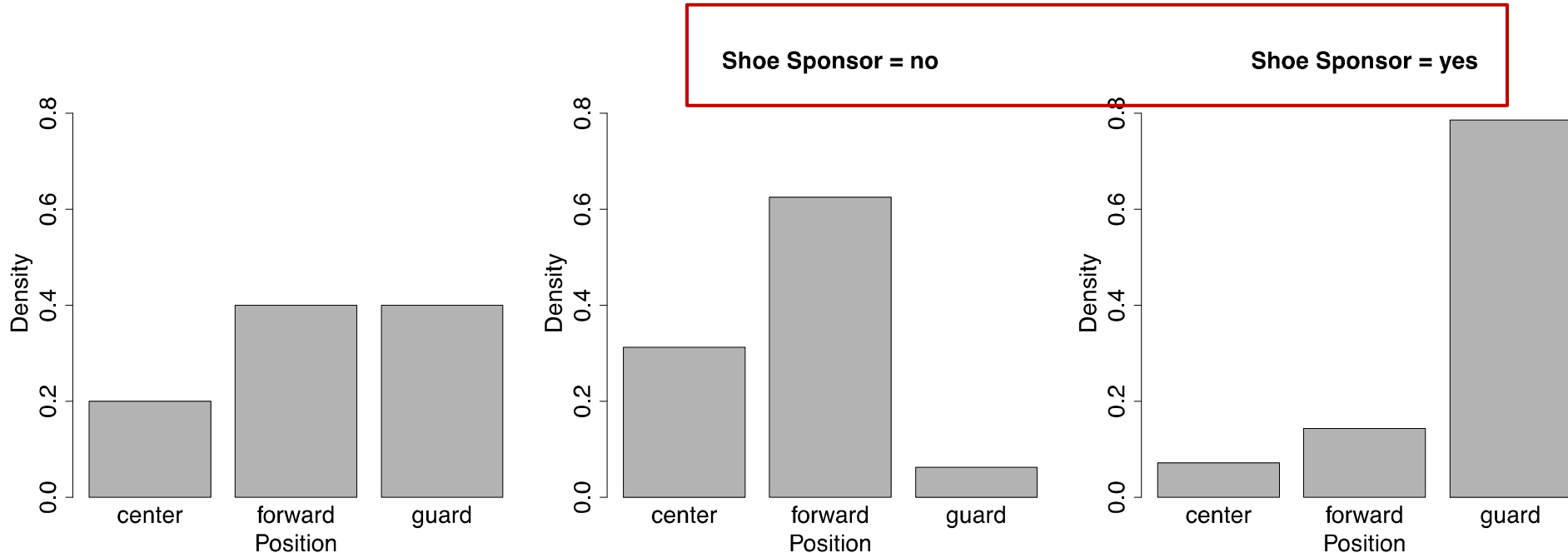
- A **scatter plot matrix (SPLOM)** shows scatter plots for a whole collection of features arranged into a matrix
- Useful for exploring the relationships between groups of features - for example, all the **continuous features** in an ABT can be shown in one SPLOM (hard to read if showing more than 10 features)

# Scatter Plot Matrix



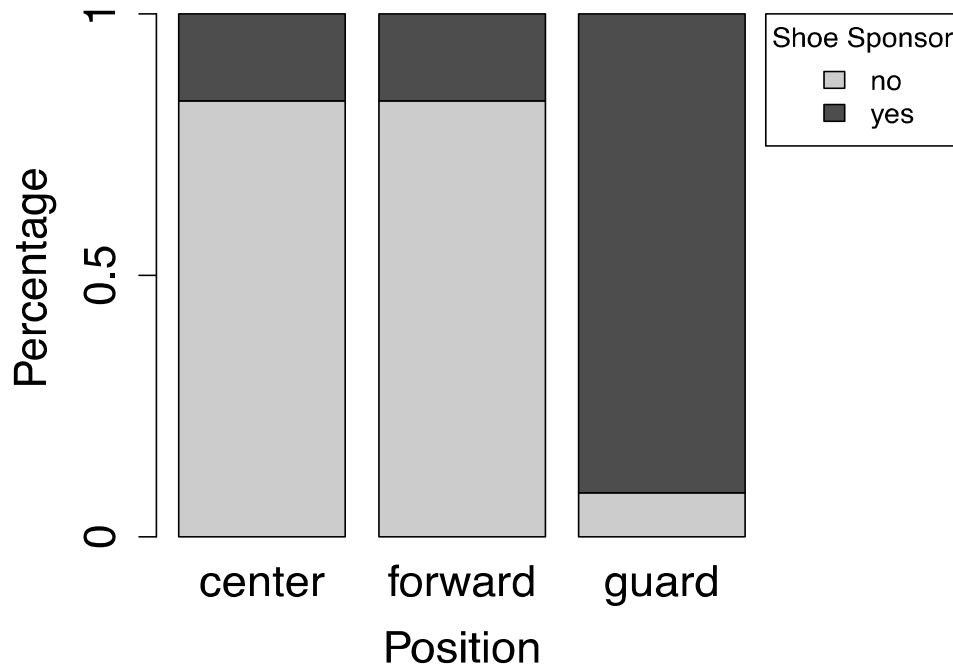
# Categorical vs Categorical: Bar Plots

- To visualize the relationship between **two categorical features** we can use a collection of small multiple bar plots
- **Position vs Shoe Sponsor** (middle and left plots)



# Stacked Bar Plots

- If the number of levels of one of the features being compared is not too high, we can use **stacked bar plots** as an alternative to the small multiples approach



| ID | Position | Height | Weight | Sponsorship Earnings | Shoe Sponsor | Career Stage | Age |
|----|----------|--------|--------|----------------------|--------------|--------------|-----|
| 1  | forward  | 192    | 218    | 561                  | yes          | veteran      | 29  |
| 2  | center   | 218    | 251    | 60                   | no           | mid-career   | 35  |
| 3  | forward  | 197    | 221    | 1312                 | no           | rookie       | 22  |
| 4  | forward  | 192    | 219    | 1359                 | no           | rookie       | 22  |
| 5  | forward  | 198    | 223    | 362                  | yes          | veteran      | 29  |
| 6  | guard    | 166    | 188    | 1536                 | yes          | rookie       | 21  |
| 7  | forward  | 195    | 221    | 694                  | no           | veteran      | 25  |
| 8  | guard    | 182    | 199    | 1678                 | yes          | rookie       | 21  |
| 9  | guard    | 189    | 199    | 385                  | yes          | mid-career   | 27  |
| 10 | forward  | 205    | 232    | 1416                 | no           | rookie       | 24  |
| 11 | center   | 206    | 246    | 314                  | no           | mid-career   | 29  |
| 12 | guard    | 185    | 207    | 1497                 | yes          | rookie       | 23  |
| 13 | guard    | 172    | 183    | 1383                 | yes          | rookie       | 24  |
| 14 | guard    | 169    | 183    | 1034                 | yes          | rookie       | 24  |
| 15 | guard    | 185    | 197    | 178                  | yes          | mid-career   | 29  |
| 16 | forward  | 215    | 232    | 434                  | no           | mid-career   | 30  |
| 17 | guard    | 158    | 184    | 162                  | yes          | veteran      | 29  |
| 18 | guard    | 190    | 207    | 648                  | yes          | mid-career   | 27  |
| 19 | center   | 195    | 235    | 481                  | no           | mid-career   | 28  |
| 20 | guard    | 192    | 200    | 427                  | yes          | mid-career   | 32  |
| 21 | forward  | 202    | 220    | 542                  | no           | mid-career   | 31  |
| 22 | forward  | 184    | 213    | 12                   | no           | mid-career   | 32  |
| 23 | forward  | 190    | 215    | 1179                 | no           | rookie       | 22  |
| 24 | guard    | 178    | 193    | 1078                 | no           | rookie       | 21  |
| 25 | guard    | 185    | 200    | 213                  | yes          | mid-career   | 31  |
| 26 | forward  | 191    | 218    | 1855                 | no           | rookie       | 19  |
| 27 | center   | 196    | 235    | 47                   | no           | veteran      | 32  |
| 28 | forward  | 198    | 221    | 1409                 | no           | rookie       | 22  |
| 29 | center   | 207    | 247    | 1065                 | no           | veteran      | 27  |
| 30 | center   | 201    | 244    | 1111                 | yes          | mid-career   | 25  |

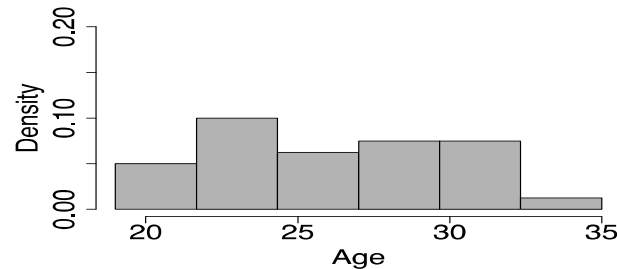
- Calculation for stacked barplot: For Position **center** 6 rows: 1 yes and 5 no (16% yes and 84% no); For Position **forward** 12 rows: 2 yes and 10 no (16% yes and 84% no); For Position **guard** 12 rows: 11 yes and 1 no (92% yes and 8% no)

# Continuous vs Categorical Features

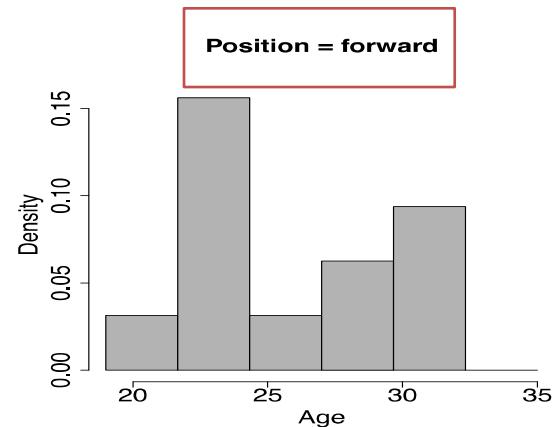
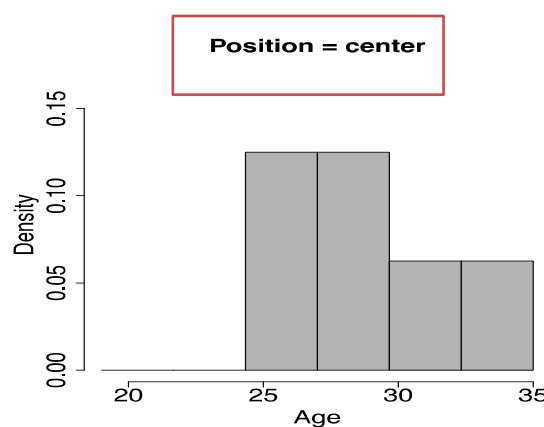
- **Small multiples** approach:
  - When we have **one continuous and one categorical feature**, for each level of the categorical feature, we plot a histogram (or a boxplot) of the values of the continuous feature

# Continuous vs Categorical Features

Example: **Age (continuous)** over all rows or broken down for **subsets of rows** for each level of feature **Position**



(a) Age



(b) Age and Position

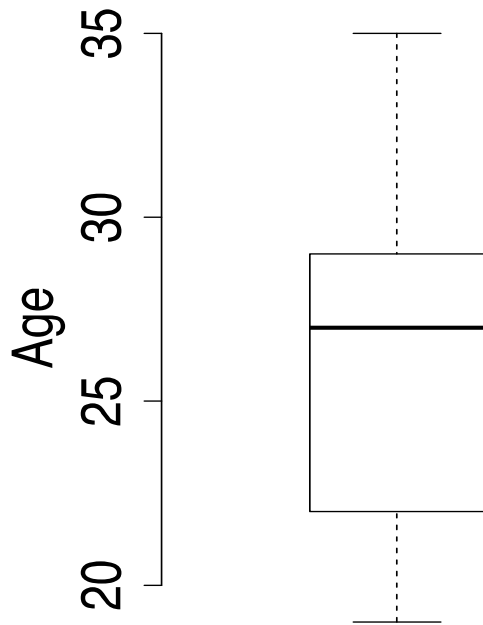


# Continuous vs Categorical Features

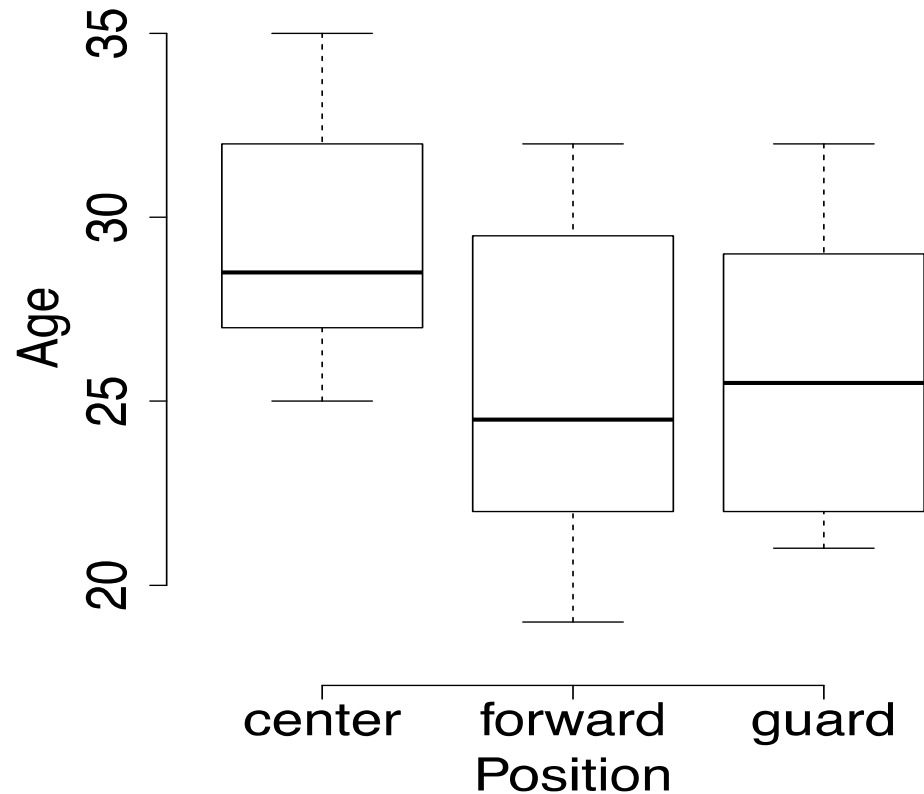
- **Small multiples** approach:
  - Instead of histograms, we can use a collection of **box plots**
- For each level of the categorical feature, plot a box plot of the corresponding values of the continuous feature

# Continuous vs Categorical Features

- Age overall and for different values of Position



(a) Age



(b) Age and Position

# Data Understanding

- Visualizing Relationships Between Features
- **Measuring Covariance & Correlation** (numeric measures to describe feature-pair relationship strength)

# Covariance of Two Features

As well as visually inspecting scatter plots, we can calculate formal measures of the relationship between two continuous features using **covariance** and **correlation**.

For two features,  $a$  and  $b$ , in a dataset of  $n$  instances, the **sample covariance** between  $a$  and  $b$  is

$$\text{cov}(a, b) = \frac{1}{n-1} \sum_{i=1}^n ((a_i - \bar{a}) \times (b_i - \bar{b})) \quad (1)$$

where  $a_i$  and  $b_i$  are values of features  $a$  and  $b$  for the  $i^{\text{th}}$  instance in a dataset, and  $\bar{a}$  and  $\bar{b}$  are the sample means of features  $a$  and  $b$ .

# Covariance of Two Features

- **Covariance values are unbounded:** fall into the range  $[-\infty, \infty]$ 
  - negative values indicate a negative relationship, values near zero indicate that there is little or no relationship between the features, positive values indicate a positive relationship
- Hard to interpret strength of relationship with the covariance, so we turn to **normalized covariance**, aka **correlation**

# Correlation of Two Features

**Correlation** is a normalized form of covariance that ranges between  $-1$  and  $+1$ .

The correlation between two features,  $a$  and  $b$ , can be calculated as

$$\text{corr}(a, b) = \frac{\text{cov}(a, b)}{\text{sd}(a) \times \text{sd}(b)} \quad (2)$$

where  $\text{cov}(a, b)$  is the covariance between features  $a$  and  $b$  and  $\text{sd}(a)$  and  $\text{sd}(b)$  are the standard deviations of  $a$  and  $b$  respectively.

# Correlation of Two Features

- **Correlation values fall into the range  $[-1, 1]$ :**  
values close to  $-1$  indicate a very strong **negative correlation**;  
values close to  $1$  indicate a very strong **positive correlation**; values around  $0$  indicate **no correlation**
- Features that have no correlation are said to be **independent**
- This definition of correlation is also called **linear correlation** or **Pearson correlation** (there are other types of correlation, e.g., Spearman correlation)

# Pearson Correlation

What is the Pearson Correlation of vectors  
[1,2,3,4,5] and [10,20,30,40,50]?



# Correlation Matrix

- **Correlation matrix** shows the correlation between each pair of features

$$\text{correlation matrix}_{\{a,b,\dots,z\}} = \begin{bmatrix} \text{corr}(a, a) & \text{corr}(a, b) & \cdots & \text{corr}(a, z) \\ \text{corr}(b, a) & \text{corr}(b, b) & \cdots & \text{corr}(b, z) \\ \vdots & \vdots & \ddots & \vdots \\ \text{corr}(z, a) & \text{corr}(z, b) & \cdots & \text{corr}(z, z) \end{bmatrix}$$

# Correlation Matrix

- Correlation matrix for the **Height** feature and the **Weight** and **Age** features from the basketball players dataset
- Matrix is symmetric around diagonal (it is enough to **read only the upper or the lower triangle** above/below diagonal)

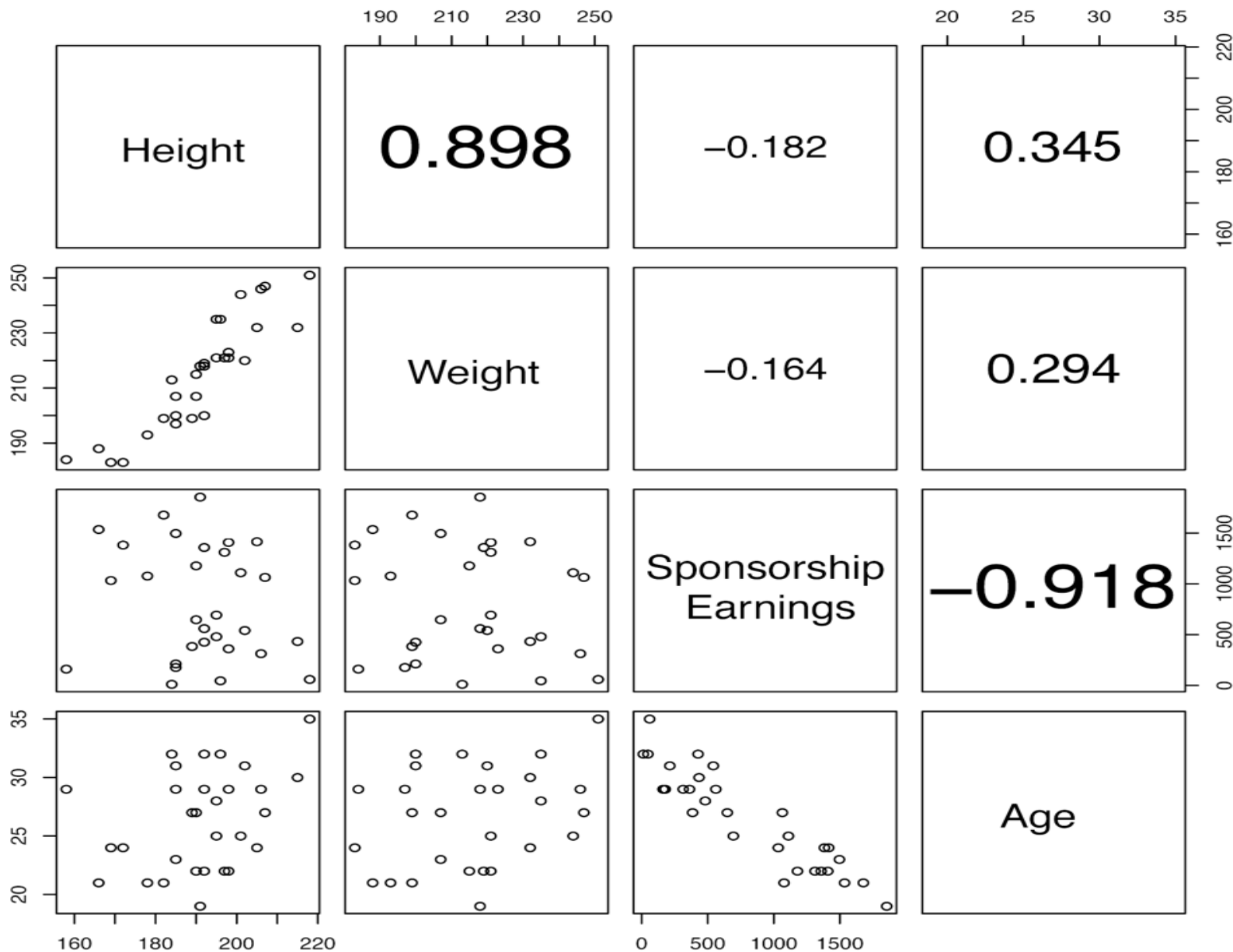
correlation matrix  $\equiv$   $\langle \text{Height}, \text{Weight}, \text{Age} \rangle$

|        | Height | Weight | Age   |
|--------|--------|--------|-------|
| Height | 1.0    | 0.898  | 0.345 |
| Weight | 0.898  | 1.0    | 0.294 |
| Age    | 0.345  | 0.294  | 1.0   |

# Correlation Matrix

- The **scatter plot matrix** (SPLOM) is a visualization of the correlation matrix
- This can be made more obvious by including the correlation coefficients in SPLOMs in the cells above the diagonal
- SPLOM is symmetrical around the diagonal:  
 **$\text{corr}(\text{Height}, \text{Weight}) = \text{corr}(\text{Weight}, \text{Height})$**

# Scatter Plot & Correlation Matrix



# Scatter Plot & Correlation Matrix

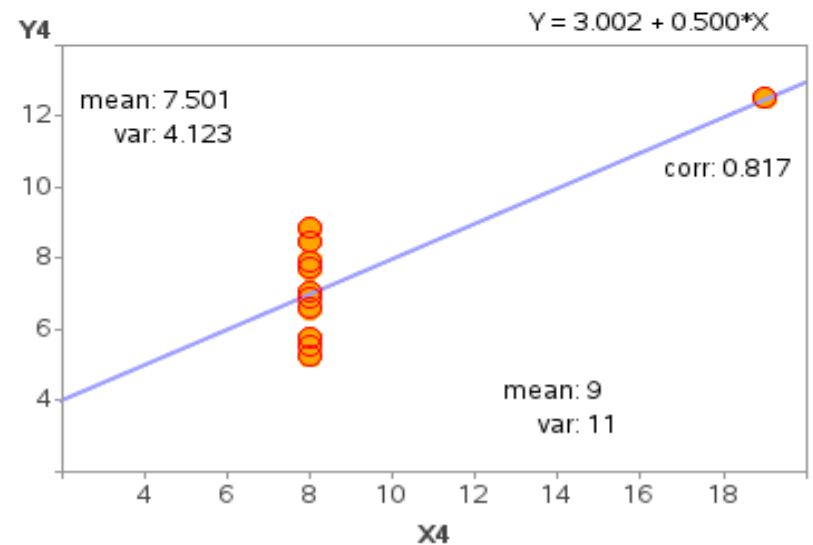
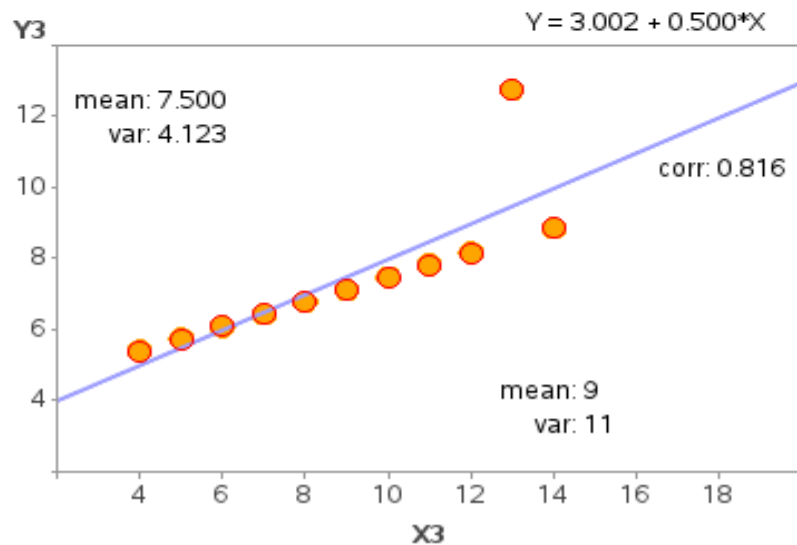
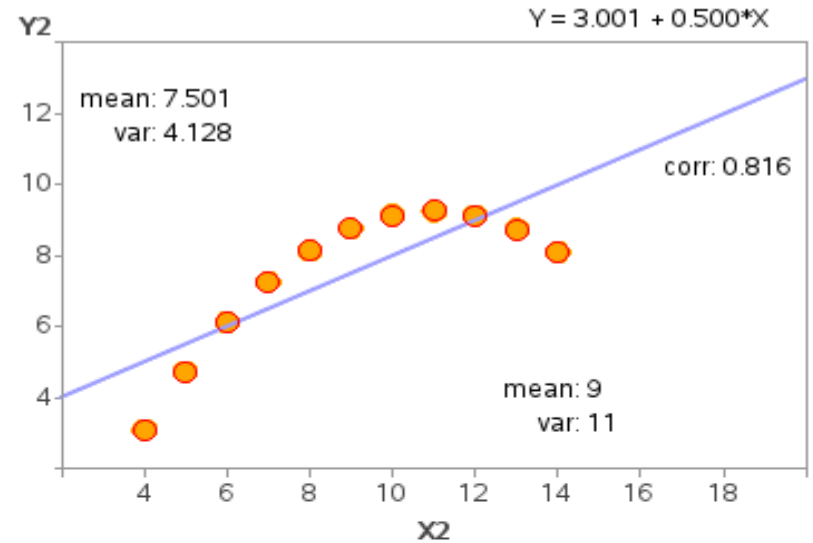
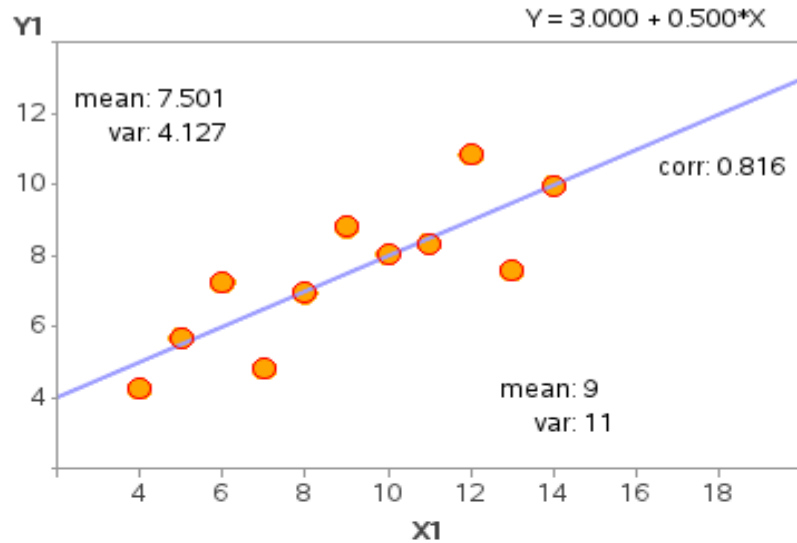
- **Linear (Pearson) correlation** is a useful measure of the relationship between two continuous features, but not always appropriate
- Some **limitations of linear correlation** are illustrated very clearly in the famous example **Anscombe's quartet** by **Francis Anscombe** (**key take-away:** summary statistics don't tell the whole story; we need to also plot the data to really understand what is going on)
- Linear correlation can be replaced with alternatives (e.g., Spearman Rank Correlation used for non-linear correlation:  
[https://en.wikipedia.org/wiki/Spearman%27s\\_rank\\_correlation\\_coefficient](https://en.wikipedia.org/wiki/Spearman%27s_rank_correlation_coefficient) )

# Anscombe's Quartet

4 different sets of data with identical correlation coefficients (.81).  
What does this say about each data set? Not much.

| data set 1              |       | data set 2 |      | data set 3 |       | data set 4 |      |
|-------------------------|-------|------------|------|------------|-------|------------|------|
| X1                      | Y1    | X2         | Y2   | X3         | Y3    | X4         | Y4   |
| 10                      | 8.04  | 10         | 9.14 | 10         | 7.46  | 8          | 6.58 |
| 8                       | 6.95  | 8          | 8.14 | 8          | 6.77  | 8          | 5.76 |
| 13                      | 7.58  | 13         | 8.74 | 13         | 12.74 | 8          | 7.71 |
| 9                       | 8.81  | 9          | 8.77 | 9          | 7.11  | 8          | 8.84 |
| 11                      | 8.33  | 11         | 9.26 | 11         | 7.81  | 8          | 8.47 |
| 14                      | 9.96  | 14         | 8.1  | 14         | 8.84  | 8          | 7.04 |
| 6                       | 7.24  | 6          | 6.13 | 6          | 6.08  | 8          | 5.25 |
| 4                       | 4.26  | 4          | 3.1  | 4          | 5.39  | 19         | 12.5 |
| 12                      | 10.84 | 12         | 9.13 | 12         | 8.15  | 8          | 5.56 |
| 7                       | 4.82  | 7          | 7.26 | 7          | 6.42  | 8          | 7.91 |
| 5                       | 5.68  | 5          | 4.74 | 5          | 5.73  | 8          | 6.89 |
| .81                     |       | .81        |      | .81        |       | .81        |      |
| Correlation Coefficient |       |            |      |            |       |            |      |

# Anscombe's Quartet



# References

- **FMLPDA Book: Fundamentals of Machine Learning for Predictive Data Analytics**, John D. Kelleher, Brian Mac Namee, Aoife D'Arcy, MIT Press, 2015 ([machinelearningbook.com](http://machinelearningbook.com))
- <https://www.autodesk.com/research/publications/same-stats-different-graphs>